

Technical Development Roadmap

Engineering & Product Delivery Plan

Prepared for	Executive Leadership, Investors, Stakeholders
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Table of Contents

1. Project Overview
2. Development Timeframe Estimation
3. Team Composition
4. Module-wise Breakdown
5. Phase-wise Delivery Plan
6. Technical Architecture Recommendations
7. Timeframe & Team Size Summary
8. Risk Analysis
9. Final Recommendation

1. Project Overview

1.1 Platform Description

Haddiya is a luxury multi-vendor gift e-commerce platform headquartered in the Sultanate of Oman. The platform unifies premium gifting categories — perfumes, fine jewelry, watches, flowers, handbags, heritage items, chocolates, and children’s products — into a single mobile-first, bilingual (Arabic / English, RTL-aware) marketplace.

Haddiya differentiates itself from generic marketplaces by combining a glassmorphism luxury UI, an AI-powered Gift Assistant (audience → occasion → budget recommendation flow), curated send-as-gift workflows (message cards, scheduled delivery, gift wrapping), and a comprehensive vendor self-service portal.

1.2 Business Objectives

#	Objective	KPI
1	Capture share of GCC premium gifting market	GMV growth ≥ 20% QoQ
2	Onboard curated vendor base	150 vendors Y1, 500 Y2
3	Drive repeat purchase via personalization	Repeat-buyer rate ≥ 35%
4	Enable cross-border GCC delivery by V2	International GMV ≥ 25%
5	Maintain luxury brand perception	NPS ≥ 60, app rating ≥ 4.7

1.3 Target Users & Stakeholders

Primary user segments

- Affluent consumers (25–55 yrs) gifting for family, corporate, social occasions.
- Expatriate professionals in GCC sending gifts to local recipients.
- SMB corporate buyers (bulk gifting for events, employee recognition).

Vendor segment

- Boutique luxury retailers, florists, perfumers, jewelers, artisanal producers.
- Single-vendor and multi-branch operations.

Internal stakeholders

- Platform Operations (catalog quality, vendor onboarding, dispute resolution).
- Marketing & Growth (campaigns, promo codes, segmentation).
- Finance (settlement, commission, payouts).
- Customer Success (order tracking, refunds, gift-card support).

1.4 Major System Components

Layer	Components
Consumer Channels	iOS App, Android App, Mobile Web (PWA), Desktop Web
API Gateway / BFF	Auth, rate-limiting, localization, routing
Core Commerce	Catalog, cart, pricing, orders, payments, shipping, wishlist, checkout
Vendor Portal	Onboarding, inventory, orders, earnings
Admin CMS	Ops, catalog moderation, reports, vendor management

Layer	Components
Intelligence Layer	Gift AI Assistant, recommendations, search ranking, promotion engine
Platform Foundation	PostgreSQL, Redis, OpenSearch, S3, SQS, CDN, WAF

2. Development Timeframe Estimation

Estimates assume a dedicated, co-located product team with stable scope. All durations expressed in calendar weeks and months. Buffer (~15%) included.

Release	Scope	Weeks	Months	Notes
MVP	Core consumer flow, single payment gateway, 1 language, basic vendor portal	14–16	3.5–4	Launch-ready in Oman, 20–30 pilot vendors
V1 Production	Full bilingual, multi-payment, Gift AI v1, vendor portal, admin CMS, push	26–30	6.5–7.5	Public launch, marketing-ready
V2 Enterprise	Multi-country GCC, advanced AI, loyalty, B2B gifting, analytics, multi-warehouse	44–52	11–13	Regional expansion, enterprise SLAs

Total cumulative duration (sequential): ~52 weeks / 12–13 months.

Recommended parallelization: MVP → V1 overlap saves 6–8 weeks.

End-to-end calendar with overlap: ~10–11 months to V2-ready state.

3. Team Composition

3.1 Recommended Team (V1 Production Build)

Role	HC	Engagement	Duration	Notes
Product Manager / Product Owner	1	Full-time	Full project	Roadmap, vendor liaison
Project Manager / Scrum Master	1	Full-time	Full project	Ceremonies, delivery, risk
UX Researcher	1	Part-time (50%)	Phases 1–2	Bilingual user testing
UI/UX Designer	2	Full-time	Phases 1–4	Lead + product designer
Frontend Engineers (Web)	2	Full-time	Phases 3–7	React/Next.js, RTL/i18n
Mobile Engineers	2	Full-time	Phases 3–7	React Native / native
Backend Engineers	3	Full-time	Phases 3–7	NestJS or Go
AI/ML Engineer	1	FT Ph5, PT after	Months 4–8	Gift Assistant, embeddings
DevOps / Platform	1	Full-time	Phases 3–7	IaC, CI/CD, observability
QA Lead	1	Full-time	Phases 4–7	Test strategy, automation
QA Engineers	2	Full-time	Phases 5–7	Manual + Playwright, bilingual
Security Engineer	1	Part-time (40%)	From Ph3	Pen testing, OWASP, PCI
Total active FTEs (peak)	~15	—	—	Plus part-time PM/PO/Sec

3.2 Lean MVP Team Variant

Role	HC	Engagement
Product Owner / PM (dual-hat)	1	Full-time
UI/UX Designer	1	Full-time
Full-stack Engineers	3	Full-time
Mobile Engineer	1	Full-time
QA Engineer	1	Full-time
DevOps (shared/fractional)	1	Part-time (40%)
Total	~7	—

3.3 Enterprise V2 Team Variant

Scale to ~22–25 FTEs: add a second AI engineer, data engineer, SRE, customer success engineer, dedicated mobile pod per platform (iOS + Android native), and an analytics engineer.

4. Module-wise Breakdown

Effort expressed in person-weeks (PW). Complexity reflects technical risk, not size alone.

#	Module	Description	Cmplx	Deps	P W	Resources
1	Authentication & Authorization	Email/password, phone+OTP, Apple, Google. RBAC.	High	—	10	1 BE, 1 FE, 1 Sec
2	User Management & Profiles	Profile, addresses, preferences, GDPR controls.	Medium	Auth	6	1 BE, 1 FE
3	Vendor Onboarding & Mgmt	KYC upload, approval, store config, payout setup.	High	Auth, Doc	14	1 BE, 1 FE, ops
4	Catalog Management	Multi-vendor CRUD, variants, media, bilingual content.	High	Vendor Mgmt	16	1.5 BE, 1 FE
5	Inventory & Stock	Per-SKU, per-warehouse stock, reservations, alerts.	Medium	Catalog	8	1 BE
6	Search & Filtering	OpenSearch, facets, typo tolerance, bilingual relevance.	High	Catalog	10	1 BE, AI input
7	Cart & Wishlist	Persistent cart, vendor grouping, guest merge, sync.	Medium	Catalog, Auth	6	1 BE, 1 FE, 1 Mob
8	Checkout & Orders	Multi-step, split orders, gift wrapping, scheduling.	High	Cart, Pay, Ship	14	1.5 BE, 1 FE, 1 Mob
9	Payment Integration	Thawani/Tap, Apple Pay, COD. Tokenization, 3DS, refunds.	High	Checkout	12	1 BE, 1 Sec
10	Shipping & Scheduling	Carrier integration, zones, time windows, webhooks.	High	Checkout	10	1 BE
11	Promotions & Coupons	Coupon engine, stacking rules, vendor vs platform.	Medium	Checkout	6	1 BE, 1 FE
12	Notifications	Push (FCM/APNs), in-app, email, WhatsApp templates.	Medium	Auth, Orders	8	1 BE, 1 Mob
13	Gift AI Assistant	Guided flow + LLM + embeddings hybrid ranking.	High	Catalog, Search	12	1 AI, 1 BE
14	Recommendation Engine	Collaborative filtering, FBT, category affinity.	High	Catalog, Orders	10	1 AI, 1 BE
15	Vendor Portal	Dashboard, products, orders, earnings, payouts.	High	Vendor Mgmt	18	1 BE, 1.5 FE
16	Admin / Ops Panel	Approvals, moderation, banners, disputes, refunds.	High	All core	16	1 BE, 1 FE
17	Reporting & Analytics	GMV, AOV, cohort retention, funnel. DW + dashboards.	High	All transactional	12	1 Data, 1 BE
18	Document Management	Vendor KYC, invoices, gift card art, S3 signed URLs.	Medium	Auth	5	1 BE
19	Loyalty & Points	Points ledger, earn/redeem rules, tiering.	Medium	Orders, Pay	8	1 BE, 1 FE
20	Customer Support	Ticket flow, chat handoff, refund processing.	Medium	Orders	6	1 BE, 1 FE

#	Module	Description	Cmplx	Deps	P W	Resources
21	Mobile Apps	RN-shared, deep-linking, biometrics, push, offline cart.	High	All consumer APIs	24	2 Mobile
22	Security & Compliance	OWASP, PCI, PDPL, WAF, rate-limit, secrets.	High	Cross-cutting	10	1 Sec, DevOps
23	DevOps & Platform	IaC, CI/CD, environments, observability, on-call.	High	Cross-cutting	14	1 DevOps

Total: ~253 person-weeks.

5. Phase-wise Delivery Plan

Phase 1 — Discovery & Planning · Weeks 1–3

- Stakeholder interviews, vendor persona workshops.
- Functional + non-functional requirements register.
- Information architecture, ERD, service decomposition map.
- Cloud region & vendor selection (AWS me-central-1 / Bahrain).
- Compliance scoping: Oman PDPL, PCI-DSS SAQ scope, VAT rules.

Deliverables: PRD, system architecture doc, ADRs, initial Jira backlog.

Phase 2 — UI/UX Design · Weeks 2–6 (overlapping)

- User research with 6–10 Omani consumers + 4–6 vendors.
- Wireframes → design system → high-fidelity Figma screens.
- Interaction prototypes for checkout, Gift AI, send-as-gift.
- Accessibility audit (WCAG 2.2 AA targets).

Deliverables: Figma library, design tokens JSON, motion specs, Lottie assets.

Phase 3 — Backend Development · Weeks 4–18

- Postgres schema implementation, migrations.
- Service skeletons: auth, catalog, cart, order, payment, vendor.
- API contracts (OpenAPI 3.1), generated FE/Mobile clients.
- Background workers (BullMQ / SQS) for emails, inventory, payouts.
- Integration with Thawani sandbox, Aramex test API.

Deliverables: Versioned APIs, contract tests, seed data, staging environment.

Phase 4 — Frontend & Mobile · Weeks 8–22

- Component library on top of design system (Storybook).
- Web app (Next.js 14 App Router), PWA shell.
- Mobile app (React Native with native modules).
- State management (TanStack Query + Zustand).
- i18n with full RTL support, OMR currency + Arabic dates.

Deliverables: Browsable web app + signed iOS/Android beta builds.

Phase 5 — AI & Automation · Weeks 14–24

- Product embedding pipeline (OpenAI/Cohere → pgvector).
- Gift Assistant conversational flow with structured-output guarantees.
- Recommendation API: hybrid CF + content + business-rule boost.
- Promo automation, expiring-cart nudges.
- Vendor inventory anomaly detection.

Deliverables: AI service API, evaluation harness, A/B harness.

Phase 6 — Testing & QA · Weeks 16–28 (continuous)

- Unit tests: $\geq 70\%$ coverage critical paths.
- Integration tests: Postman/Pact contract tests across services.
- E2E: Playwright (web) + Detox (mobile) for top 25 journeys.
- Bilingual QA matrix (LTR + RTL $\times$ iOS + Android $\times$ AR + EN).
- Performance: k6 load testing 5 $\times$ expected peak.
- Security: SAST (Semgrep), DAST (ZAP), third-party pen test.
- UAT with pilot vendor cohort.

Deliverables: Test reports, defect closure log, sign-off documents.

Phase 7 — Deployment & DevOps · Weeks 18–30

- IaC: Terraform modules per environment (dev, staging, prod).
- CI/CD: GitHub Actions $\rightarrow$ ECR $\rightarrow$ ECS Fargate / EKS, blue-green.
- Observability: CloudWatch + Datadog or Grafana, OpenTelemetry.
- Logging: structured JSON to OpenSearch, retention policies.
- Backups: PITR for RDS, daily S3 versioning, cross-region.
- Incident response runbooks, on-call rotation, status page.

Deliverables: Production environment, runbooks, DR drill report.

6. Technical Architecture Recommendations

Layer	Recommendation	Rationale
Frontend (Web)	Next.js 14 (App Router) + TypeScript + Tailwind v4	SSR/ISR for SEO, mature i18n + RTL
Mobile	React Native (Expo bare), native modules for biometrics	Single team, ~80% code reuse
Backend	Node.js (NestJS, TS); Go for hot paths	Type-safe full stack, large hiring pool
API Gateway / BFF	AWS API Gateway + GraphQL BFF or REST + tRPC	Tailored payloads per channel
Database	PostgreSQL 16 on RDS (multi-AZ). pgvector for embeddings.	ACID for orders, JSON for attributes
Caching	Redis (ElastiCache)	Sub-ms latency, sessions/cart/rate-limit
Search	OpenSearch with Arabic + English ICU analyzers	Faceted, typo-tolerant, RTL
Queue / Events	Amazon SQS + EventBridge (Kafka V2 optional)	Decouples checkout, notifications, payouts
Object Storage	Amazon S3 + CloudFront CDN	Images, KYC, gift card art, signed URLs
Cloud Region	AWS me-central-1 (UAE) primary, eu-south-1 DR	Data residency for PDPL
CDN & Edge	CloudFront + Lambda@Edge	Image optimization, Arabic font subsetting
Identity	AWS Cognito or Auth0; WhatsApp Business OTP	Offload undifferentiated auth
Payments	Thawani (OMR), Tap, Apple Pay, COD	Local rails first, expand later
Observability	Datadog (APM + logs + RUM) or Grafana Cloud	Full-stack visibility
Error Tracking	Sentry (web + mobile + backend)	First-class RN + Next.js integration
Feature Flags	LaunchDarkly or Unleash (OSS)	Safe rollouts, vendor segment targeting
Analytics	Segment → Snowflake/BigQuery → Looker / Metabase	Vendor dashboards, cohort analysis
Secrets	AWS Secrets Manager + Parameter Store	Rotation, audit logging
WAF / Security	AWS WAF + Shield, Cloudflare optional	Bot, DDoS, OWASP top-10

7. Timeframe & Team Size Summary

Consolidated view of delivery duration and developer headcount required to complete each release tier of the Haddiya platform.

7.1 Delivery Timeframe

Tier	Weeks	Months	Outcome
Lean MVP	14–16 weeks	3.5–4 months	Launch-ready in Oman with 20–30 pilot vendors
V1 Production	26–30 weeks	6.5–7.5 months	Public launch, bilingual, marketing-ready
V2 Enterprise	44–52 weeks	11–13 months	GCC expansion, enterprise SLAs
End-to-end (overlapped)	~44 weeks	~10–11 months	Kickoff → V2-ready with parallel phases

7.2 Developers Required per Tier

Tier	Total HC	Engineers	Designers	QA	PM/PO	DevOps/Sec
Lean MVP	~7	4 (3 FS + 1 Mob)	1	1	1 dual-hat	0.4
V1 Production	~13–15	8 (3 BE + 2 FE + 2 Mob + 1 AI)	2 + 0.5 UX	3 (Lead + 2)	2	1 + 0.4 Sec
V2 Enterprise	~22–25	18 (5 BE + 3 FE + 4 Mob + 2 AI + 2 Data + 2 SRE)	3	5 (Lead + 4)	4	2 + 1 Sec

7.3 Detailed V1 Engineering Breakdown

Role	Developers	Engagement	Duration
Backend Engineers	3	Full-time	7 months
Frontend Web Engineers	2	Full-time	7 months
Mobile Engineers (React Native)	2	Full-time	7 months
AI / ML Engineer	1	Full-time Ph5, part later	5 months
DevOps / Platform Engineer	1	Full-time	7 months
QA Lead	1	Full-time	6 months
QA Engineers	2	Full-time	6 months
UI / UX Designers	2	Full-time Ph 1–4, part later	6 months
UX Researcher	1	Part-time (50%)	2 months
Product Manager	1	Full-time	7 months
Project Manager / Scrum Master	1	Full-time	7 months
Security Engineer	1	Part-time (40%)	5 months
Peak Total Headcount	~15 FTE	—	7 months

7.4 Recommended Pod Structure (V1)

- **Consumer Pod** — 2 FE + 2 Mobile + 1 BE + 1 Designer. Owns web, mobile, checkout, search.
- **Marketplace Pod** — 2 BE + 1 FE + 1 Designer. Owns vendor portal, admin CMS, catalog.
- **Platform Pod** — 1 AI/ML + 1 DevOps + 1 Sec (PT). Owns Gift AI, infra, security.
- **QA Pod** — 1 Lead + 2 QA. Bilingual coverage across all pods.

8. Risk Analysis

8.1 Technical Risks

Risk	Likelihood	Impact	Mitigation
RTL/Arabic edge cases in third-party libs	High	Medium	Bilingual QA matrix from sprint 1
Multi-vendor order split complexity	High	High	DDD, saga pattern, contract tests
Payment gateway certification timelines	Medium	High	Phase 1 start, sandbox-first, COD fallback
Mobile app store review delays	Medium	Medium	TestFlight early, metadata hygiene
AI hallucinations / off-brand recs	Medium	Medium	Constrained generation, catalog grounding

8.2 Scalability Concerns

Concern	Trigger	Mitigation
Database write contention on order spikes	> 100 orders/min	Read replicas, sharding, back-pressure
Search index latency under filter-heavy load	Catalog > 50K SKUs	OpenSearch capacity, query caching
Mobile push fan-out (campaigns)	> 100K devices	SQS-backed batch, rate-aware workers
Image bandwidth load	High mobile traffic	Adaptive CDN, AVIF/WebP, responsive sizes
Vendor catalog import bursts	New vendor cohort	Async import pipeline, validation queue

8.3 Security Risks

Risk	Severity	Mitigation
PII exposure (Oman PDPL, GDPR)	Critical	Encryption at rest (KMS) + TLS 1.3, field-level enc
Payment data scope	Critical	Tokenization via PSP, no PAN storage, SAQ-A scope
Account takeover (credential stuffing)	High	Rate limiting, fingerprinting, MFA, breach check
Vendor portal privilege escalation	High	Strict RBAC, per-vendor isolation, tenant scoping
Supply-chain (npm/PyPI) attacks	Medium	Dependabot, SBOM, Snyk/Socket, signed images
Document upload abuse (KYC malware)	Medium	AV scan, file-type validation, S3 isolation

8.4 Performance Bottlenecks

Bottleneck	Symptom	Fix
N+1 queries on product listing	Slow category pages	DataLoader/batched joins, read models
Cold-start on serverless	First-request latency	Provisioned concurrency or ECS containers
Cart recompute per render (mobile)	Janky cart UI	Memoized totals, optimistic updates

Bottleneck	Symptom	Fix
Large bundle on web	Slow LCP on 3G	Code-splitting, dynamic imports, Brotli
Synchronous payout calculations	Slow checkout confirmation	Async settlement, eventual reconciliation

9. Final Recommendation

9.1 Ideal Team Size — Three Options

Team size scales with delivery speed. Same V1 scope can be delivered with a smaller team over a longer timeline, or a larger team over a shorter one. Three viable configurations:

Option	Team Size	V1 Timeline	Best For	Tradeoff
Lean (Startup)	8–9 FTE	8–9 months	Budget-conscious founders, single-product focus	Sequential phases, no parallel pods, founders wear multiple hats
Standard (Recommended)	13–15 FTE	6.5–7.5 months	Funded company targeting on-time public launch	Balanced velocity vs. coordination overhead; full feature parity
Aggressive	20+ FTE	5–6 months	Race-to-market situations, well-funded	Diminishing returns; coordination tax; higher onboarding cost

Honest take: 14 FTE feels large only if compared to MVP-only scope. For full V1 — bilingual web + iOS + Android + vendor portal + admin CMS + payment integrations + Gift AI — it is right-sized. If budget is the constraint, the Lean 8–9 FTE option ships the same scope in 8–9 months instead of 7.

- **Pod structure (Standard option):** Consumer Web/Mobile · Vendor Portal + Admin · Platform/AI · QA.
- **Pod structure (Lean option):** Single full-stack pod + 1 designer + 1 QA + fractional DevOps.

9.2 Best Timeline for Delivery

Milestone	Calendar Week	Outcome
Kickoff + Discovery	W 0–3	PRD, architecture frozen
Design + Backend foundations begin	W 4	Figma library v1, API contracts
MVP private beta	W 16	Pilot vendors transacting
V1 public launch (Oman)	W 30	Marketing-ready, app stores live
V2 GCC expansion	W 52	UAE, KSA, Bahrain coverage

Total: ~12 months from kickoff to V2-ready state with overlap.

9.3 MVP Feature Set (Ship First)

Must-have in MVP:

- Auth: phone+OTP, social login.
- Bilingual catalog (AR + EN) with categories, search.
- Cart, wishlist, single-vendor + multi-vendor checkout.
- Address book, delivery scheduling (date + time slot).
- Single payment gateway (Thawani) + COD.
- Send-as-gift: message card, scheduled delivery, recipient details.
- Vendor portal: onboarding, product CRUD, order list, basic earnings.
- Order tracking, push + in-app notifications.
- Admin: vendor approvals, basic catalog moderation.
- Promo code engine (single-stack coupons).

9.4 Defer to V1 / V2

Defer to V1:

- Full Gift AI Assistant (rule-based simple in MVP, LLM-powered in V1).
- Loyalty / points ledger.
- Advanced analytics dashboards.
- Recommendation engine v1.
- WhatsApp Business notifications.
- Multi-payment gateway redundancy.

Defer to V2:

- Multi-country expansion (UAE, KSA, Bahrain) with multi-currency, multi-warehouse.
- B2B corporate gifting portal (bulk orders, invoicing, custom branding).
- Subscription gifting / recurring deliveries.
- Advanced ML personalization (home feeds).
- Vendor mobile app.
- Marketplace ads / sponsored placement.
- Live chat / video consultations with vendors.

9.5 Fastest Realistic Approach (Without Quality Compromise)

- **Freeze MVP scope hard.** No additions without scope-swap. Feature flags so half-done work doesn't block release.
- **Parallelize Phase 2 (design) with Phase 3 (backend foundations).** Saves 4–6 weeks of calendar time.
- **Adopt monolith-modular backend first** (NestJS modules), extract microservices only when scaling forces it.
- **Use React Native** with high code reuse instead of dual native apps — saves ~3 months and 1.5 FTE.
- **Buy, don't build, undifferentiated components:** Auth0/Cognito, OpenSearch managed, Sentry, Datadog.
- **Continuous deployment from day one.** Every PR deployable to staging; production deploys daily once V1 stabilizes.
- **Vendor pilot cohort (10–15 vendors) on MVP** while V1 features land — real-world feedback loop.
- **Dedicated bilingual QA from Phase 4.** Arabic/RTL bugs caught early cost 10× less than post-launch.
- **Pre-book third-party penetration test** at Week 24 so findings land before public launch.
- **Reserve 15% capacity for unplanned work.** Vendor requests, regulatory clarifications, payment quirks.

Bottom line:

A focused team of ~14 engineers, designers, QA, and product staff delivers production-ready V1 in ~7 months. Alternatively, a lean 8–9 FTE team ships the same V1 scope in ~8–9 months — the right choice for budget-conscious startups. Either path scales to enterprise V2 over the following ~6 months by growing the team to ~22–25 FTE.

Document prepared for executive review. Detailed sprint-level backlog, capacity plan, and risk register available as appendices upon request.